



Mohammad Yusuf

Master of Technology
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RESEARCH INTERESTS

Machine Learning on Graphs, with applications in scientific discovery, neuroscience, and complex systems.

EDUCATION

Degree	Institute	CGPA/Percentage	Year
M.Tech., Computer Technology	Indian Institute of Technology Delhi, New Delhi	8.88/10	2023 - 2025
B.Tech., Computer Engineering	Aligarh Muslim University, Aligarh	9.38/10	2019 - 2023
Senior Secondary	Saiyyid Hamid Sr. Sec. School, AMU Aligarh	89.2%	2019
Secondary	Montfort Sr. Sec. School, Roorkee	10/10	2017

EXPERIENCES

- Graduate Teaching Assistant** Jan 2024 - Dec 2024
Indian Institute of Technology Delhi New Delhi, India
 - Tutored undergraduate students in "Computer Architecture" and "Intro to Electrical Engineering Lab" courses for two semesters.
 - Marked exam papers, provided constructive feedback, and guided students on assignments. Conducted lab sessions weekly to help students in experiments and clear doubts during office hours.
- Indian Institute of Technology Roorkee** June 2022 - July 2022
Summer Research Intern Roorkee, India
 - The topic of the research was an Application of Dynamic Graphs in Pangenomics, titled Incremental Computation of Safe Flow Path under the supervision of [Dr. Shahbaz Khan](#).
 - The problem was looked at from both a theoretical and a experimental point of view. The theoretical part looked at how safe paths change when there is an update, and the experimental part looked at heuristics using the properties of these variations to improve the practical performance on the tested graph instances.

PROJECTS

- Brain Signal-Based Graph Neural Networks for Neurological Disorders** Jan 2024 - July 2025
Masters Thesis
 - Graph ML projects part of M.Tech. Thesis under the supervision of [Dr. Sandeep Kumar](#).
 - BrainADNet: This project utilizes functional MRI data from the REST-meta-MDD dataset to construct brain graphs for patients with major depressive disorder, enhancing graph learning models by integrating decorrelation loss, implementing skip connections, augmenting data with demographic features, and utilizing explainability methods to analyze gender-specific brain regions associated with depression.
 - RESOLVE: In this project we develop a contrastive learning-based GNN model that integrates functional connectivity patterns from both time and frequency domains, employs a reconstruction objective to preserve original connectivity, and incorporates a fairness-aware regularization term to mitigate distributional discrepancy among demographics.
- Deep Learning : Neural Networks** Aug 2024 - Dec 2024
Deep Learning course project
 - This project is part of Advances in Deep Learning (ELL891, IITD) course project under the supervision of [Dr. Sumeet Agarwal](#).
 - Implemented fundamental deep learning components from scratch, including an automatic differentiation engine, CNN layers (convolution, max-pooling, batch normalization), RNNs with BPTT, and a multihead attention mechanism for transformers.
 - Developed NLP applications such as Twitter sentiment analysis using RNNs and text summarization with a Seq2Seq model leveraging GloVe embeddings and Bi-LSTM.

• Stauffer Grimson Background Subtraction

Jan 2024 - Feb 2024

Machine Learning course project



- This project is part of Introduction to Machine Learning (ELL784, IITD) course project under the supervision of [Prof. Sumantra Dutta Roy](#).
- Implemented a Python-based background subtraction algorithm for real-time video analysis, using Gaussian Mixture Models to adapt to environmental changes and effectively isolate moving objects for applications like surveillance and traffic monitoring.

• Network Analysis in Table Tennis Championships

Aug 2023 - Nov 2023

Social Network Analysis course project



- This project is part of Social Network Analysis (ELL880, IITD) course project under the supervision of [Prof. Sougata Mukherjea](#).
- This project models and analyzes a network of table tennis players based on match data from 2013-23, including major tournaments. It verifies if the network follows real-world properties, calculates key metrics, detects communities, and ranks players using PageRank and HITS algorithms, comparing these rankings with official ITTF rankings.

• Dark Web Traffic Detection and Characterization system

Aug 2022 - May 2023

Bachelors Thesis



- Machine learning project part of B.Tech. Thesis under the supervision of [Mr. Mohammed Abdul Qadeer](#).
- Developed a graph-based approach to detect multi-layer encrypted dark web traffic, addressing the challenge of adversaries using VPN and Tor together.
- Constructed feature and adjacency matrices from CIC-Darknet-2020 network flow data, achieving 98.65% accuracy in classifying benign vs. darknet traffic.

PUBLICATIONS

- Barman, J., **Yusuf, M.**, Kumar, S. & Gandhi, T. K. (2025). Enhancing Depression Diagnosis with Augmented Brain Signal Driven Decorrelated Graph Neural Networks. *Nature Communications Medicine* (under review)

SKILLS

- **Programming Languages**: C, C++, Python, Java.
- **Machine Learning**: PyTorch, TensorFlow, Scikit-learn.
- **Assembly Languages**: 8085, 8086 and RISC-V.
- **Web Development**: React.JS, Express.JS, MongoDB, Node.JS
- **Tools**: LaTeX, Logisim, Gephi, Git, MS Office.

ACHIEVEMENTS

- Department Rank 3 out of a total of 24 students in M.Tech.
- Recieved AICTE GATE Scholarship to pursue M.Tech. in Computer Technology at IIT Delhi over a period of 2 years (2023-25).
- [All India Rank 1210, Graduate Aptitude Test in Engineering \(GATE\), CS-IT 2023, 98.40%ile](#)
- [All India Rank 4212, Graduate Aptitude Test in Engineering \(GATE\), CS-IT 2022, 94.54%ile](#)

EXTRA-CURRICULAR

- **Volunteer Organizer**, Indo-US Workshop on 6G Technologies, IIT Delhi. Organized and oversaw speaker sessions, event logistics, and participant engagement for a global academic workshop on cutting-edge 6G technologies.

CERTIFICATIONS

- **TOEFL iBT**: R (25), L (25), S (23), W (28), Total = 101/120, [ETS](#) Nov 2024
- **Deep Learning with PyTorch : Generative Adversarial Network**, [Coursera](#) Sept 2022
- **Python Data Structures**, [Coursera](#) Sept 2020